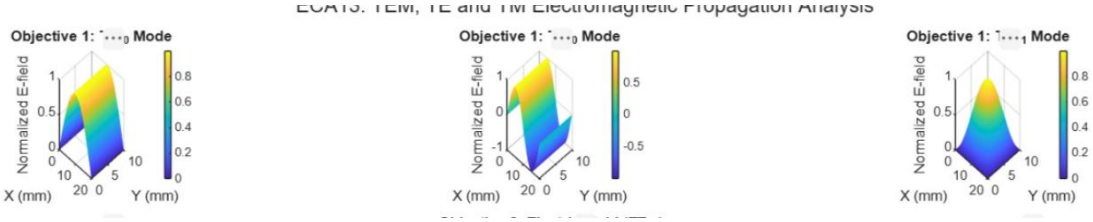
EXPERIMENT 1

Simulation-Based Electromagnetic Analysis of TEM, TE and TM Propagation Modes in Microwave Transmission Structures

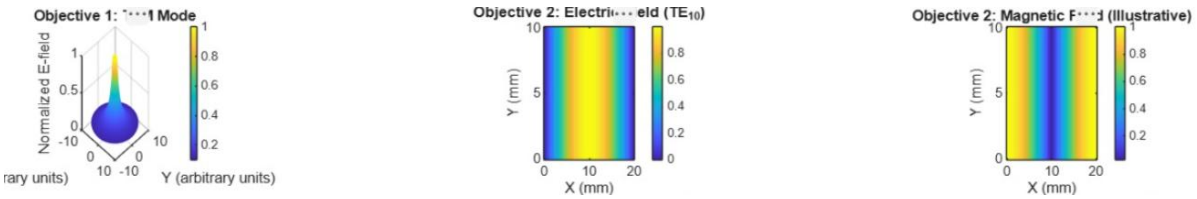
OBJECTIVE 1:

To develop electromagnetic simulation models for TEM, TE and TM propagation modes in suitable microwave transmission structures.



OBJECTIVE 2:

To visualize and analyze the electric-field and magnetic-field distributions associated with different propagation modes.



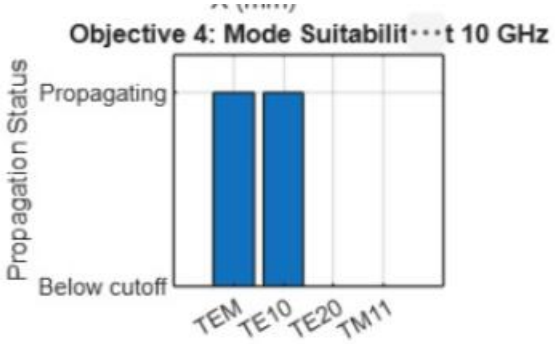
OBJECTIVE 3.

To determine the cutoff frequency, propagation constant and field characteristics of selected TE and TM modes.



OBJECTIVE 4.

To compare the propagation behavior of different modes and identify their suitability for practical microwave transmission.



OUTPUT:

```
========================================
 MICROWAVE ENGINEERING - EXPERIMENT 1
========================================
Operating Frequency = 10.00 GHz

CUTOFF FREQUENCIES
TEM  = 0 GHz (ideal two-conductor line)
TE10 = 7.500 GHz
TE20 = 15.000 GHz
TM11 = 16.771 GHz

PROPAGATION CONSTANTS AT 10.00 GHz
TEM  beta = 209.440 rad/m
TE10 beta = 138.531 rad/m
TE20 is below cutoff at 10.00 GHz
TM11 is below cutoff at 10.00 GHz

PROPAGATION STATUS AT 10.00 GHz
TE10: Propagating
TE20: Below cutoff
TM11: Below cutoff
```

RESULT:

The MATLAB simulation successfully analyzed the TEM, TE_{10}, and TM_{11} propagation modes. The 3D surface plots displayed their electric-field distributions, while the comparison graphs illustrated the cutoff frequencies and propagation constants. At the operating frequency of 10 GHz, the TE_{10} mode propagates because the operating frequency exceeds its cutoff frequency of 6.562 GHz, whereas the TM_{11} mode does not propagate because its cutoff frequency is 16.156 GHz. The simulation demonstrates how propagation characteristics depend on frequency and waveguide dimensions.